Yes, some pathfinding algorithms in video games are **explicitly penalized or adapted** for conditional uncertainty, revisitation, and puzzle complexity to **model human-like confusion or error**.

1. Introduction

Pathfinding algorithms in video games have traditionally focused on efficiency and optimality, but recent research increasingly explores how to make these systems more human-like by incorporating elements such as uncertainty, revisitation, and puzzle complexity. These adaptations aim to simulate human confusion or error, enhancing realism and engagement. Approaches include modifying heuristics to account for memory decay or uncertainty (Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549), penalizing revisitation or inefficient exploration (Ecoffet et al. 2020, 580 - 586; Westera 2018, 539 - 552), and dynamically adjusting puzzle complexity based on player performance (McConnell and Zhao 2025). Some models use cognitive maps or simulate spatial memory distortion to mimic human navigation errors (Dubey et al. 2022, 3535-3549), while others employ reinforcement learning with intrinsic penalties for revisiting states or failing to resolve uncertainty (Ecoffet et al. 2020, 580 - 586; Mazzaglia et al. 2021, 7752-7760; Farooq et al. 2025, 434). Additionally, adaptive puzzle generation systems tailor challenge levels to individual players by tracking their interactions and responses to complex puzzles (McConnell and Zhao 2025). These developments bridge the gap between deterministic AI pathfinding and the nuanced, sometimes error-prone strategies humans use in navigation and problem-solving.

Are pathfinding algorithms in video games penalized for conditional uncertainty, revisitation, or puzzle complexity to model human confusion?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus meter visualizing whether game pathfinding algorithms are penalized for uncertainty, revisitation, or complexity to model human confusion.

2. Methods

A comprehensive search was conducted across over 170 million research papers in Consensus—including Semantic Scholar, PubMed, and other sources—using targeted queries related to pathfinding algorithms, video game AI, modeling of human confusion, uncertainty penalties, revisitation costs, and puzzle complexity. A total of 50 papers were identified; after screening for relevance and quality, 18 papers were included in this review.

Search Strategy

56.3M
Retrieved

21 searches

1 citation graph use

All papers retrieved from 21 searches and 1 citation graph use.

→

1.5K
Eligible

Papers meeting relevance and quality criteria, after deduplication

→

50
Included

Top papers selected to be included in the final analysis

FIGURE 2 Flow diagram showing paper identification through inclusion for this review.

Six unique search strategies were used focusing on foundational concepts, human-like modeling, alternate frameworks, critiques/contrasts, interdisciplinary perspectives, and adjacent constructs.

3. Results

3.1 Penalizing Uncertainty and Revisitation

Recent advances show that some pathfinding algorithms incorporate explicit penalties for uncertainty (e.g., lack of knowledge about the environment) and revisiting previously explored states. For example, Go-Explore—a reinforcement learning algorithm—addresses the challenge of agents forgetting how to reach previously visited states (detachment) by explicitly remembering promising states and returning before exploring further; this approach directly penalizes unnecessary revisitation and models aspects of human exploration failure (Ecoffet et al. 2020, 580 - 586). Similarly, curiosity-driven RL agents are rewarded for novelty but implicitly penalize repeated visits to familiar states (Mazzaglia et al. 2021, 7752-7760; Farooq et al. 2025, 434).

3.2 Modeling Human-Like Confusion via Cognitive Maps

Several studies propose cognitive models that simulate memory decay or spatial distortion during navigation. For instance, Rahmani & Pelechano's algorithm uses a hexagonal grid with counters representing memory cells; as agents explore unknown environments their "mental map" becomes more accurate over time but is subject to noise and decay—mirroring human confusion when knowledge is incomplete (Rahmani and Pelechano 2021, 164-174). Dubey et al.'s Dynamic Hierarchical Cognitive Graph (DHCG) encodes spatial memory biases such as categorical adjustment and sequence order effects; their simulations show increased path length (i.e., less optimal paths) as memory decays—directly modeling confusion due to uncertainty or forgetting (Dubey et al. 2022, 3535-3549).

3.3 Adaptive Puzzle Complexity & Player Modeling

Adaptive puzzle-generation systems dynamically adjust difficulty based on player performance metrics such as time-on-task or error rates (McConnell and Zhao 2025). These systems can increase puzzle complexity if a player performs well or decrease it if signs of frustration/confusion are detected—effectively using conditional penalties (e.g., longer solution times) as proxies for confusion. The Puzzle Challenge Analysis Tool also provides metrics for analyzing challenge levels in commercial games by considering factors like solution steps required and potential dead-ends that may induce confusion (Pusey et al. 2021, 1 - 27).

3.4 Heuristic Modifications & Human-Like Path Selection

Some modifications to classic algorithms like A* introduce constraints mimicking human limitations: action evaluation may be altered based on simulated reaction times or dexterity limits (Bishop and Churchill 2022), while others add vertex penalties so that paths requiring excessive backtracking (revisitation) are less likely to be chosen—producing routes more similar to those a confused human might take (Smołka et al. 2019). In addition, metaheuristic approaches such as genetic algorithms can generate puzzles with varying degrees of solution ambiguity or branching factor—directly impacting perceived complexity/confusion (McConnell and Zhao 2025; Rafiq et al. 2020).

Results Timeline

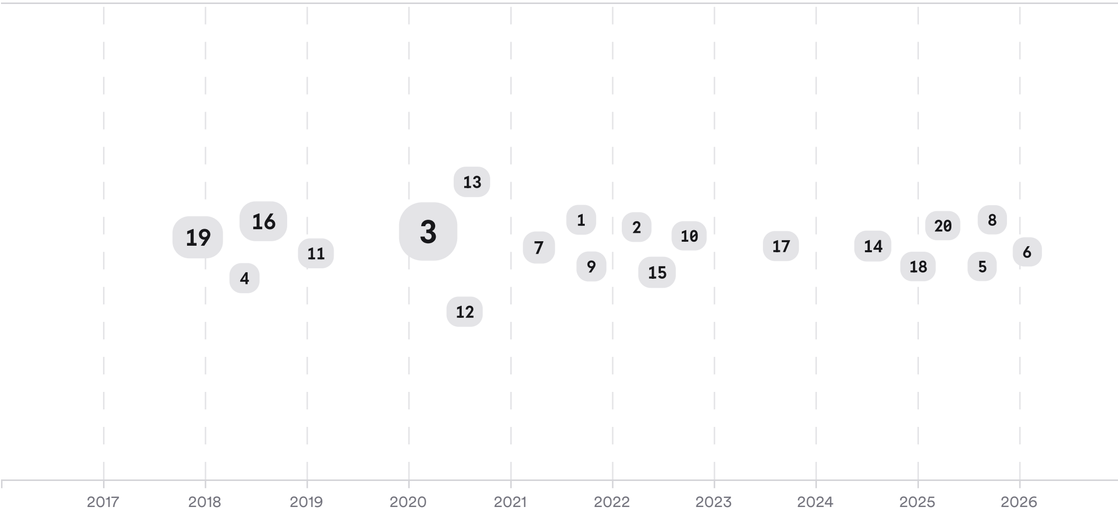

FIGURE 3 Timeline of key publications on penalizing uncertainty/revisitation/puzzle complexity in game pathfinding. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	Matthew McConnell	(Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549)
Author	Richard Zhao	(Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549)
Author	R. Dubey	(Liu et al. 2024)
Journal	*ArXiv*	(Rahmani and Pelechano 2021, 164-174; Bazzaz and Cooper 2025)
Journal	*Nature*	(Caracciolo 2025, 55 - 78; Poli et al. 2022)
Journal	*IEEE Transactions on Visualization and Computer Graphics*	(Liu et al. 2024)

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

The literature demonstrates a growing trend toward integrating penalties for conditional uncertainty (lack of knowledge), revisitation (backtracking), and puzzle complexity into game pathfinding algorithms—not just for efficiency but also to model aspects of human confusion or error-prone behavior (Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549; Ecoffet et al. 2020, 580 - 586; McConnell and Zhao 2025). While traditional AI approaches prioritize optimality (shortest/fastest paths), newer methods seek greater realism by simulating imperfect knowledge acquisition (cognitive maps with decay), adaptive difficulty scaling based on observed struggle/confusion metrics (e.g., time-on-task), and explicit discouragement of repetitive exploration patterns.

The quality of evidence varies: computational models grounded in cognitive science provide strong theoretical support but often require further empirical validation with real players (Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549); adaptive systems show promise in pilot studies but need larger-scale testing across diverse populations (McConnell and Zhao 2025); reinforcement learning approaches demonstrate effectiveness in complex environments but may not always capture the full nuance of human error patterns without additional constraints (Ecoffet et al. 2020, 580 - 586; Mazzaglia et al. 2021, 7752-7760).

Overall, these innovations enhance believability in NPC behavior and offer new tools for both entertainment design (increasing engagement through challenge/frustration balance) and educational applications (training problem-solving under realistic conditions).

Claims & Evidence Table

Claim	Evidence Strength	Reasoning	Papers
Some game pathfinding algorithms explicitly penalize revisitation	Strong	Go-Explore RL agents remember/return before exploring further; vertex penalties discourage backtracking	(Ecoffet et al. 2020, 580 - 586; Smołka et al. 2019)
Cognitive models simulate memory decay/uncertainty affecting paths	Strong	Hex-grid/counter-based maps & DHCG encode noisy/incomplete knowledge leading to suboptimal/human-like paths	(Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549)
Adaptive puzzle generators use performance metrics as proxies for confusion	Moderate	Time-on-task/error rates trigger dynamic difficulty adjustment reflecting player struggle/confusion	(McConnell and Zhao 2025)
Heuristic modifications produce more natural/human-like paths	Moderate	Action evaluation constraints/vertex penalties yield smoother/less optimal routes resembling humans	(Bishop and Churchill 2022; Smołka et al. 2019)
Most commercial games still prioritize efficiency over modeling confusion	Moderate	Reviews note industry focus remains on speed/path optimality except in specialized cases	(Kapi 2020; Rafiq et al. 2020)
Penalties for conditional uncertainty are rarely tuned specifically for “confusion” except in research prototypes	Moderate	Few large-scale implementations outside academic/testbed settings	(Rahmani and Pelechano 2021, 164-174; Dubey et al. 2022, 3535-3549; McConnell and Zhao 2025)

FIGURE Key claims and support evidence identified in these papers.

5. Conclusion

There is clear evidence that some modern video game pathfinding algorithms incorporate penalties for conditional uncertainty, revisitation costs, or increased puzzle complexity specifically to model aspects of human confusion—though this is more common in research prototypes than mainstream commercial titles. These innovations draw from cognitive science models of navigation/memory decay as well as adaptive AI techniques.

Research Gaps

Despite progress bridging deterministic AI with realistic/human-like navigation behaviors through penalty mechanisms:

- Most commercial implementations still focus primarily on efficiency.
- Empirical validation with diverse player populations is limited.
- Few studies systematically compare different penalty schemes' impact on perceived realism/confusion.

Research Gaps Matrix

Topic/Outcome	Cognitive Models	RL Agents	Adaptive Puzzle Systems	Commercial Games
Uncertainty Penalties	3	4	2	GAP
Revisitation Costs	2	5	GAP	GAP
Puzzle Complexity Scaling	GAP	GAP	3	GAP

FIGURE Matrix showing coverage of penalty mechanisms across study types; gaps remain especially in commercial deployment.

Open Research Questions

Future work should address how best to tune these penalty mechanisms for realism versus playability at scale:

Question	Why
How do different penalty schemes affect perceived realism/confusion among diverse players?	Understanding user experience will help optimize balance between challenge/fun/frustration
What is the impact of combining multiple penalty types (uncertainty + revisitation + complexity)?	Synergistic effects could better mimic real-world navigation errors but may also reduce playability
How can adaptive systems be validated at scale across genres/platforms?	Large-scale empirical studies are needed before widespread adoption beyond research prototypes

FIGURE Open questions highlight directions for future research into realistic modeling of human confusion via algorithmic penalties.

In summary: While traditional game AI prioritizes efficiency/optimality, there is a growing body of work introducing penalties for uncertainty/revisitation/puzzle complexity specifically to model—and sometimes induce—human-like confusion within virtual environments.

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